Problem Set - 10

1 Design the following counters:

(a) A mod-7 asynchronous counter using J-K FFs

(b) A mod-12 asynchronous counter using J-K FFs

(c) A mod-9 synchronous counter using J-K FFs

(d) A mod-7 synchronous counter using S-R FFs

(e) A mod-12 synchronous counter using T FFs

(f) A mod-14 synchronous counter using D FFs

2 Design a J-K counter that goes through states 2, 4, 5, 7, 2, 4,…… Is the counter self- starting? Modify the circuit such that whenever it goes to any invalid sate it comes back to state 2.

3 Design the following types of counters:

(a) A type D counter that goes through states 0, 2, 4, 6, 0.... The undesired states must always go to a 0 on the next clock pulse.

(b) A type T counter that goes through states 6, 3, 7, 8, 2, 9, 1, 12, 14, 0, 6, 3 . . .

Is the counter self starting?

(c) A type T counter that goes through states 0, 5, 4, 2, 0….. Is the counter self starting?

(d) A type T counter that must go through states 0, 2, 4, 6, 0 . . . if the control line is HIGH, and through states 0, 4, 2, 7, 0 . . . if the control line is LOW.

(d) Type T counter that must go through states 0, 2, 4, 6, 0 . . . if the control line is high and through states 0, 4, 2, 7, 0 . . . if the control line is low.

4 Design a direct logic circuit to generate the sequences 100111 and 011101 simultaneously.

5 Generate the pulse train 100110 using indirect logic.

6 Design a BCD up/ down counter using S-R FFs

7 Design an up/down counter using D FFs to count 0, 3, 2, 6, 4, 0….